

PROJECT REPORT

CYBERTHREATFORGE

*Advanced Cybersecurity Simulation, Threat Modeling,
and Defensive Operations Platform
(Amroha-Cyber-security-)*

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1. Executive Summary

CyberThreatForge (Amroha-Cyber-security-) is an enterprise-grade cybersecurity simulation, training, and threat intelligence platform designed to model complex cyber threat surfaces. The system integrates advanced threat predictive engines with post-quantum security modeling, SCADA system verification, deepfake media detection, and cyber-insurance risk analytics to form a unified defensive operations center.

The primary goal of the platform is to allow defensive security operators and investigators to simulate attack scenarios, test and audit detection systems, perform linguistic and cryptographic forensics, and validate model safeguards in real-time. By bridging the gap between machine-learning threat intelligence and simulated red/blue team exercises, CyberThreatForge delivers proactive resilience across modern digital infrastructure.

2. System Architecture & Tech Stack

Decoupled Multi-tier Microservice Architecture:

- Frontend Dashboard: Built with React 18 and TypeScript (Vite), offering responsive charting, collaborative workspaces, and administrative portals.
- Backend Services: Powered by Python (FastAPI/Flask) utilizing SQLAlchemy, SQLite database, and Alembic migrations.
- AI Analytics Layer: Custom prediction engine to classify threats, analyze zero-days, watermark models, and analyze forensic evidence.
- Infrastructure: Containerized via Docker and orchestrated with Kubernetes deployment manifests.

3. Core Security Modules & API Specifications

Module Name	Primary Functionality	REST API Endpoint
InsureGuard	Cyber risk premium forecasting	/api/insureguard/policy
ModelDefender	Model watermark embedding	/api/modeldefender/watermark
FirmwareGuard	Firmware integrity verification	/api/firmwareguard/firmware
PsyOpsForge	Misinformation tracking	/api/psyopsforge/campaigns
ZeroDayForge	Zero-day risk prediction	/api/zerodayforge/predict
DeFiGuard	Crypto transaction audits	/api/defiguard/analyze
LegacyShield	SCADA protocol validation	/api/legacyshield/investigate

4. Deployment & Orchestration Guide

Docker Compose Setup:

```
$ docker-compose up --build
```

Kubernetes Commands:

```
$ kubectl apply -f k8s/deployment.yaml
```

```
$ kubectl apply -f k8s/service.yaml
```

```
$ kubectl apply -f k8s/ingress.yaml
```